Acoustic-Linguistic Multimodal Representation for Speech Classification.

1. Hypotheses

- Acoustic-Linguistic Fused Architecture performs better than Acoustic-Only, or Linguistic-Only models.
- ASR errors affect downstream classification negatively.
- The model degrades as the size of the training data decreases, i.e, additional information is useful.

2. Experiment Table

S. No	Experiment Name	Category/Issue
1.	Audio-Only	Baseline
2.	Text-Only (Reference Transcripts	Baseline
3.	Text-Only (Whisper Transcripts)	Baseline
3.	Acoustic + Linguistic - Direct Fusion	Baseline
4.	Acoustic + Linguistic - Projected Fusion	Fusion Architecture Change
5.	Whisper Transcript vs Reference Text	Audit
6.	Hyperparameter Tuning - Early Stopping, Regularization, Weighted Classes	Tuning
7.	Training the Model at: 25%, 50%, 75% of the Data	Data Scarcity